

Metastable curled tori and their front-driven decay in combinatorial quantum gravity

Emily Smith^{1,*}

¹*Independent researcher, Charlottesville, Virginia, USA*

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Combinatorial quantum gravity assigns to a four-regular bipartite graph an energy built from its total Ollivier curvature, which splits into a global term that rewards squares and a local term that removes the reward for a third square on an edge. We treat the coefficient λ of the local term as a control parameter; $\lambda = 1$ is the published model. For $\lambda \geq 1$ the flat lattice torus is the ground state exactly, and a $4 \times L$ torus, one direction curled to a single square, lies $4(\lambda - 1)$ per vertex above it. At $\lambda = 1.25$ the curled torus is metastable and decays at fixed coupling by a memoryless nucleation followed by a single front: at half conversion 98.8 to 99.8% of vertices belong to one ordered state or the other, and the released energy is exactly $4(\lambda - 1)$ per vertex. The barrier is a single local move. In a sealed system a seed of 12 energy units starts the change at every size from 48 to 192 vertices and nothing smaller does, while the release grows with the system. Sealed, the change completes only if its surroundings can absorb the release, and it leaves one defect of fixed size however large the torus. Mapped from $\lambda = 1.05$ to 1.45, the torus is metastable up to 1.35 and the change keeps its two-order, single-front character wherever it is, finishing cleanly near $\lambda = 1$ and leaving more defects as λ grows. The release vanishes at $\lambda = 1$. Every verdict quoted was pre-registered.

I. INTRODUCTION

In combinatorial quantum gravity (CQG) [1–3] geometry emerges from random graphs driven by a discrete Ricci curvature. The question studied so far is whether geometry forms out of the random phase sharply or smoothly: a first-order transition with hysteresis when only the global term is kept [4], a continuous or hybrid one with the full action [2, 3]. A different kind of change has not, as far as we have found, been studied in this model: one *ordered* arrangement turning into another. Such changes matter for the proposal that dark matter consists of long-lived regions of the graph stuck in a different discrete arrangement [5]. We study the simplest case the model offers, a torus with one direction curled, opening into the flat torus. It exists as a metastable state only for $\lambda > 1$; we map it along λ and follow its decay.

II. MODEL AND EXACT RESULTS

States are four-regular bipartite graphs obeying the hard-core rule (no two vertices share more than two neighbours) [2, 3]. With S the number of squares, S_e the squares on edge e and $X = \sum_e (S_e - 2)_+$,

$$H = 16(N - S) + 4\lambda X = 4 \sum_e [(2 - S_e) + \lambda(S_e - 2)_+]. \quad (1)$$

At $\lambda = 1$, H is the total combinatorial curvature of Refs. [2, 3]; λ multiplies the local (saturation) term. The hard-core rule limits $S_e \leq 3$. Three exact consequences follow.

Ground state. For $\lambda \geq 1$ every term in the sum is non-negative, so $H \geq 0$, with $H = 0$ for any graph with two

squares on every edge: the flat lattice torus. For $\lambda > 1$ nothing else reaches zero.

Ladder. The $4 \times L$ torus has $S = 5N/4$ and $X = N$ (the edges of each wrap-around 4-cycle carry three squares), so $H/N = 4(\lambda - 1)$; the 4-cube has $H/N = 8(\lambda - 1)$. Each curled direction costs $4(\lambda - 1)$ per vertex (Fig. 1). We call $d(v)$, the number of pairs of edges at v that close no square, the local dimension: 2 on the flat torus, 1 on the curled torus, 0 on the 4-cube.

Metastability window. Enumerating every valid graph at $N = 18$ (26 classes up to relabelling) [8], states above the ground state from which every single move costs energy exist for $1 < \lambda < 1.6$, and at neither $\lambda = 1$ nor any $\lambda \geq 1.6$ (steps of 0.05 up to 1.7, coarser to 30).

The way out. The Markov chain swaps one partner each between two vertices of the same side (an edge switch preserving the bipartition; Metropolis; the hard-core rule is enforced) [8]. From the perfect $4 \times L$ torus the cheapest moves are of two kinds, counted exactly: move A ($\Delta S = -2$, $\Delta X = -4$; cost $32 - 16\lambda$; $3N$ moves) and move B ($\Delta S = -4$, $\Delta X = -10$; cost $64 - 40\lambda$; $2N$ moves). A sweep is $2N$ attempts drawn from $4N^2$ equally likely proposals, so A is offered 3 times and B twice per sweep, independent of N . The mean time to the first exit is therefore predicted without free parameters,

$$\tau(\lambda, g) = [3e^{-(32-16\lambda)/g} + 2e^{-(64-40\lambda)/g}]^{-1}, \quad (2)$$

with A the cheaper for $\lambda < 4/3$.

III. DECAY AT FIXED COUPLING

At $\lambda = 1.25$ the barrier is 12. Over twelve conditions ($g = 1.4$ to 2.5, $N = 64$ and 144, 16 runs each), the measured mean waiting time against the move-A term of Eq. (2) has a plain mean ratio of 0.98 and a weighted mean of 0.88 ± 0.06 , every ratio between 0.71 and 1.62,

* emilysmithcreate@gmail.com

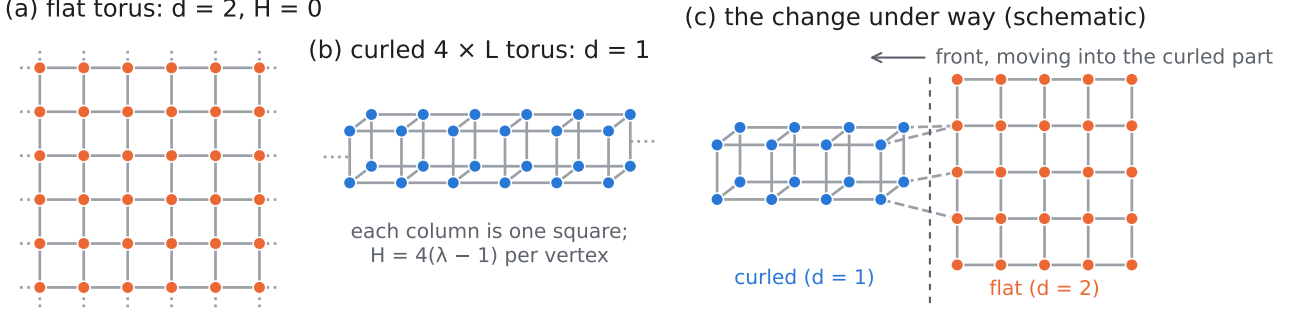


FIG. 1. (a) The flat lattice torus. (b) The $4 \times L$ torus drawn as a tube: its short direction is a single square, so the edges round each column carry three squares. (c) Schematic of the change of Sec. III, not a simulation snapshot: the curled order on one side of a single front, the flat order on the other.

over a ninety-fold range of times; a fit of the barrier at $N = 144$ gives 12.1 ± 0.9 [Fig. 2(a)]. Including move B lowers the predicted waiting time by 15% at $g = 1.5$; across λ , the full Eq. (2) describes the measured waits over two decades (Sec. V).

The decay itself was pre-registered [8]: at $\lambda = 1.25$, $g = 1.5$, thirty decays at each of $N = 64, 96, 144, 192$, and the adjacency read when the conversion fraction first crossed 25, 50 and 75%. The predictions were a memoryless waiting time (coefficient of variation between 0.7 and 1.3), at least 80% of vertices at $d \in \{1, 2\}$ at half conversion, and the converted vertices in one piece holding at least 70% of them. All three hold at every size and on two independent sets of seeds (Table I): the change starts at one place and runs as a single front, with the two orders on either side of it and almost nothing in between [Fig. 2(b)]. The mean waiting time does not fall with N , as a local barrier offered at an N -independent rate per sweep requires.

The released energy is exact. After the change, 24 to 28 of 30 decays per size rest on the flat torus with release $4(\lambda - 1)$ per vertex to within 1%. The others settle on metastable defected sheets, read from their wiring: most often one column of the torus left curled, four vertices at $d = 1$ with $S = N + 1$, $X = 6$, costing $24\lambda - 16$ (14 at $\lambda = 1.25$); more rarely combinations of small defects of 8 to 45 units. The pre-registered verdict, a two-state

change, holds at $N = 64, 96$ and 192 ; at $N = 144$ one decay changed state inside its final measuring window and the size was set aside by the energy gate as written, with all three predictions holding there too.

For contrast: at the same $\lambda = 1.25$ the change out of the random phase shows one hump in the energy distribution at every coupling, size and replica we ran, with any latent heat below 1.28 per vertex at $N = 100$ and falling with N [8]. In this model the sharp change is between orders.

IV. SEALED SYSTEM

To follow the released energy we replace the heat bath by Creutz demons [6]: C stores, each never negative; each attempted move picks one store to pay for it or be paid, and is refused if that store cannot pay. H plus the stores' contents is conserved exactly, checked to the last unit in every run, and the stores' mean reads the temperature.

A local, size-independent threshold. With one store holding a seed s and nothing else, the torus converted in 8 of 8 runs for $s \geq 12$ and in 0 of 8 for $s \leq 11.5$ (seeds 4 to 16), at each of $N = 48, 64, 96, 144, 192$ (320 runs). Twelve is the cost of move A. With nothing to spare the torus never converts, although the flat torus is lower. The cost of starting is fixed while the release, $4(\lambda - 1)N$, grows with the system.

The surroundings must absorb the release. With the seed in one of C stores ($N = 64, 96, 192$; twenty runs each; 420 runs): the converted region melts into the random phase in 17 to 20 of 20 runs for $C \leq N/16$ and in 9 to 16 of 20 at $C = N/8$; at $C = N/4$ most runs end partly converted or partly melted; for $C \geq N/2$ the flat torus results in 18 to 20 of 20. The crossover scales with N , as a release of $4(\lambda - 1)N$ into C stores against a melting coupling near 3.5 predicts ($C^* \approx N/3.5$). No run stalled with both orders present and the bath below

TABLE I. The decay of the $4 \times L$ torus at $\lambda = 1.25$, $g = 1.5$ (first run / independent rerun, thirty decays each).

N	mean wait	CV	$d \in \{1, 2\}$ at 50%	largest piece
64	871 / 660	0.91 / 0.84	0.990 / 0.998	0.98 / 0.99
96	913 / 928	1.10 / 1.02	0.990 / 0.988	0.95 / 0.96
144	856 / 743	1.01 / 0.76	0.997 / 0.997	0.92 / 0.96
192	1039 / -	0.87 / -	0.995 / -	0.85 / -

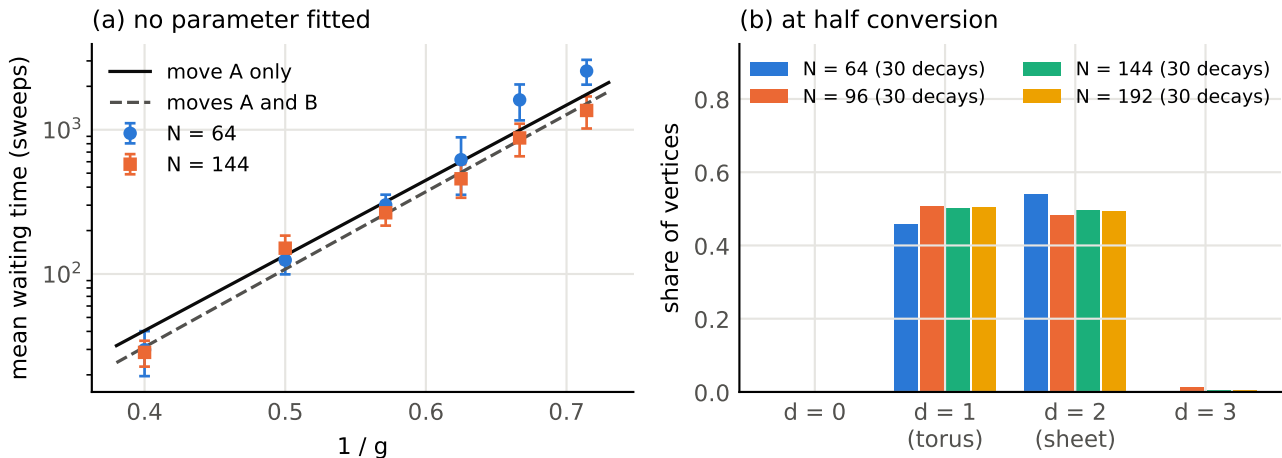


FIG. 2. (a) Mean waiting time of the perfect $4 \times L$ torus at $\lambda = 1.25$ against inverse coupling, sixteen runs per point, with Eq. (2) using move A only (solid) and moves A and B (dashed); nothing is fitted. (b) Share of vertices at each local dimension d when half the torus has converted, averaged over thirty decays per size.

the melting coupling: the curled torus has no coexistence temperature with the flat one.

One leftover, however large. In the coldest box ($C = 2N$), the number of leftover defects per converted torus is 0.90 ± 0.31 , 0.85 ± 0.37 , 1.05 ± 0.22 and 0.95 ± 0.22 at $N = 64, 96, 192, 288$ (twenty runs each; slope 0.0003 ± 0.0003 per vertex); the defect is the curled column above (14 units), occasionally a two-vertex twist (20 units). Its position relative to where the change started is indistinguishable from uniform.

V. DEPENDENCE ON λ

The decay was mapped at $\lambda = 1.05$ to 1.45 in steps of 0.05 , with the protocol of Sec. III at all four sizes and thirty decays per (λ, N) (Table II, Fig. 3). The torus counts as stuck at (λ, N) when more than half its decays are still at least 75% torus after the 200-sweep rest that defines the waiting time.

The torus is stuck from $\lambda = 1.05$ to 1.35 at every size and not at 1.40 or 1.45 ; Eq. (2) puts the mean wait at 200 sweeps near $\lambda = 1.33$. Wherever it is stuck, the change keeps the character of Sec. III: at half conversion 95 to 100% of vertices belong to one order or the other, the converted region is one piece holding 76 to 100% of it, and the release is exactly $4(\lambda - 1)$ per vertex wherever the flat torus is reached. The mean waiting time follows Eq. (2) with nothing fitted over two decades [Fig. 3(a)], and departs from it at both ends for different reasons. Near the edge the departure is in the measurement: a wait cannot be recorded before the 200-sweep watch ends (first check at 205), and for waits spread exponentially with mean τ the expected recorded mean is about $205 + \tau e^{-205/\tau}$ (dashed; also fitted to nothing): 462 at $\lambda = 1.30$ and 265 at 1.35 , against 403–505 and 280–350 measured.

Near $\lambda = 1$ the measured waits run 15 to 43% longer than Eq. (2), which the formula does not contain. Eq. (2) gives the time to the *first* exit; when the release is small, some first exits may fall back into the torus instead of growing into a front, and if so about 70 to 87% of first exits go through at $\lambda = 1.05$. That reading is tested directly by counting exits and fall-backs (PREREGISTRATION T22 in Ref. [8]). How completely the change finishes depends on λ [Fig. 3(b)]: at $\lambda = 1.05$ every decay ends at the flat torus, while the share ending on a defected sheet rises to 21% at 1.30 and 54% at 1.35 . At fixed coupling, then, the leftover is a feature of the larger coefficients, and near $\lambda = 1$ the change is clean. Beyond the edge ($\lambda \geq 1.40$) the torus falls apart everywhere at once and ends mostly as a defected sheet.

The pre-registered verdict for the map as a whole is inconclusive, and the reason is stated. Besides the coexistence and single-front criteria, which hold everywhere the torus is stuck, it required the waiting times to be memoryless, with a coefficient of variation between 0.7 and 1.3; at thirty decays that band is about two standard errors wide, and it fails in five of the 28 stuck cells (at $\lambda = 1.10, 1.30$ and 1.35). At $\lambda = 1.35$ the energy check also fails for two to five decays per size, whose final states match none of the allowed ones. A repair computed before being proposed (the variation of the time after the rest) changes no verdict.

VI. DISCUSSION

At $\lambda = 1$, the published model, the flat, curled and 4-cube tori are degenerate and the change releases nothing; the $N = 4p^2$ sizes recommended for the random-to-lattice problem admit neither of the latter two. The sharp change between two orders described here therefore

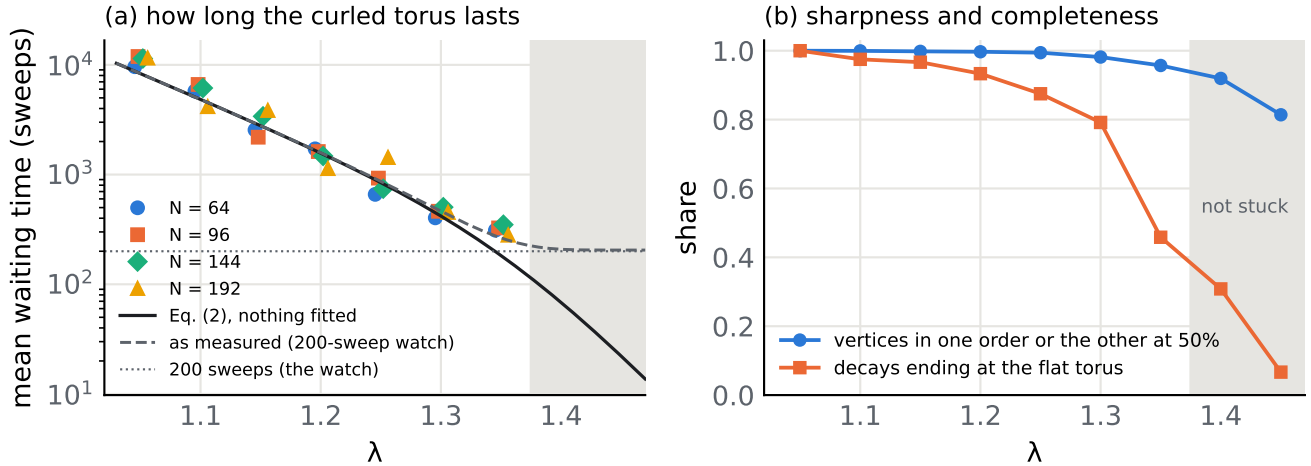


FIG. 3. The map along λ at $g = 1.5$. (a) Mean waiting time where the torus is stuck, four sizes, with Eq. (2) (solid) and the same prediction as it would be recorded given the 200-sweep watch, $205 + \tau e^{-205/\tau}$ (dashed); both fitted to nothing. Shaded, the λ at which it is not stuck. (b) Share of vertices in one order or the other at half conversion, and share of decays ending at the flat torus, pooled over sizes.

belongs to the family on the $\lambda > 1$ side; the map of Sec. V shows it keeping its character down to $\lambda = 1.05$, where it is cleanest, with a release that shrinks as $4(\lambda - 1)$ and vanishes at the published value. The same questions can be asked at $\lambda = 1$ of regions stuck in a different discrete arrangement inside a curved background [5]: whether such a region is metastable, and whether it gives way by a front with a definite release. The tools used here (exact barrier counting, front tracking, sealed baths) apply unchanged.

Not claimed: a first-order transition in the thermodynamic sense (this is the decay of a metastable state at fixed coupling, and “sharp” is used in the pre-registered sense above); anything at sizes beyond 288; why the cold box leaves exactly one defect. All runs label vertices; treating them as interchangeable [7] weights each state by its number of symmetries, which is $2N$ for the curled

torus and 2 one move away, so the escape would be slower by a factor of about N , with the release unchanged.

Outlook (motivation, not a result). This work is motivated by the hypothesis that space emerged from a specific ordered precursor through a sharp change of this kind, releasing a definite energy and leaving remnants behind. Nothing here tests that hypothesis beyond showing that a model of emergent geometry contains such a change. We note only that the remnant found here, of fixed size and one per change however large the system, is the kind of object a dark-matter remnant of such a change would have to be, and that the change needs a push of fixed size while what it releases grows with the system.

METHODS AND DATA

Simulations use single-edge switches that preserve the bipartition, with Metropolis acceptance; the chain reproduces exact averages computed by enumerating every valid graph at $N = 16$ and 18, and the published square density of Ref. [2] (Fig. 8a) at $N = 160$ to rms 0.005. Seeds are fixed in configuration files; every result file carries its configuration and software versions. Predictions and analysis rules were committed before the runs they concern, with every amendment recorded. Code, data, configurations and pre-registrations: [8].

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TABLE II. The map along λ at $g = 1.5$ (ranges over $N = 64$ to 192; thirty decays per cell; $\lambda = 1.25$ from Table I). Two orders: share of vertices at $d \in \{1, 2\}$ at half conversion. Flat: share of decays ending at the flat torus.

λ	τ , Eq. (2)	mean wait	two orders	flat
1.05	8330	9617–11915	1.000	100%
1.10	4844	4154–6626	0.998–1.000	98%
1.15	2788	2191–3834	0.996–0.999	97%
1.20	1570	1132–1716	0.996–0.998	93%
1.30	419	403–505	0.980–0.984	79%
1.35	183	280–350	0.948–0.963	46%
1.40	68	not stuck	0.911–0.924	31%
1.45	22	not stuck	0.799–0.837	7%

nel is tested against an independent computation before use. We thank C. A. Trugenberger for correspondence.

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